

done
a7

IDENTIFICATION

Product Code: MAINDEC-08-D1HA-D
Product Name: PDP-8, 8/1, Extended Memory
Address Test
Date Created: March 13, 1968
Maintainer: Diagnostic Group
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1. ABSTRACT

The PDP-8, 8/I Extended Memory Address Test tests all of memory not occupied by the program to make sure that each location can be uniquely addressed. This is performed by a series of four tests. The first two tests write the address and complement address of each memory location into itself, and then checks the contents of each location to make sure each is correct. The third test first sets all of memory not occupied by the program to all ones, and then writes a word of all zeroes, except for one bit, into each location and checks for error. The fourth test is similar except that a word of all ones, except for one bit, is written into each location and checks for error.

2. REQUIREMENTS

2.1 Equipment

A standard PDP-8 or 8/I with a minimum of 8K words of core memory.

2.2 Storage

The program requires locations 0010 to 2534 octal.

2.3 Preliminary Programs

The Binary loader must be in locations 7756-7776 octal. Also, all diagnostics for a basic PDP-8 or 8/I must have previously been run successfully.

3. LOADING PROCEDURE

3.1 Method

- a. Turn off the Teletype reader.
- b. Set the SR to 7777.
- c. Press LOAD ADDRESS, and then START.
- d. Place the Binary tape in the Teletype reader and turn on the reader.
- e. When the program has been loaded, stop the computer, turn off the reader, and remove the tape.

4. STARTING PROCEDURE

4.1 Starting Address

Start from address 200 to specify the amount of core memory to test, SR settings, and to receive a header print-out.

4.2 Restarting Address

Start from address 211 to change the test limits, SR settings, and to inhibit the header print-out.

4.3 Operator Action

Immediately after starting from address 200 or 207, the program will print "TEST LIMITS". The operator must then specify, via the Teletype keyboard, the amount of core memory to test, followed by a carriage return.

The following rules govern the amount of memory to test:

- a. Type two octal numbers, separating the numbers with a comma. The first number signifies the lowest order 4K stack to test; the second signifies the highest order.
- b. The program expects the 4K stacks to be numbered sequentially starting with stack 0.
- c. If the highest order stack to test is typed as the first stack, the program will interchange the two values so as to make the second value the first to test.
- d. After typing the second octal number, press the carriage return key to terminate the line.
- e. The program will test the lowest and highest order 4K stack specified, plus every stack between, starting with the lowest specified.
- f. Any single stack, or two or more sequential stacks may be specified.
- g. The stack containing the program may be included when specifying two or more stacks. The stack containing the program will be tested after automatic program relocation takes place (see section 5.3.1).
- h. If a typing error is made, press the RUB-OUT key. "TEST LIMITS" will be printed again. All previous input is disregarded.

For the following examples assume the program to be located in stack 0, and the program has been started from address 200 or 207. The amount of core memory available is 32K.

Example A: TEST LIMITS
0,7↵ (↵ denotes carriage return)

Example A indicates stacks 0, 1, 2, 3, 4, 5, 6, and 7 will be tested.

Example B: TEST LIMITS
7,0↵

The program will perform exactly as Example A.

Example C: TEST LIMITS
4,5 ↵

Only stacks 4 and 5 will be tested.

Example D: TEST LIMITS
3,3 ↵

Stack 3 alone will be tested.

Example E: TEST LIMITS
0,0 PROGRAM IS LOCATED IN FIELD 0
TEST LIMITS
0,1 ↵

Example E shows the message printed by the program when a single stack is selected which currently contains the program. "TEST LIMITS" is printed again, and the operator must then correct the test limits.

Operation of the program is unpredictable if the amount of memory selected for testing exceeds the actual amount available, i.e., selecting 32K for testing on a PDP-8 or 8/1 equipped with a maximum of 28K.

4.3.1 Setup SR - After the test limits is specified, the program will print "SETUP SR". For normal program operation, the SR must be set to equal 0000₍₈₎. Press the carriage return key after setting the SR to 0000. The program will then run until stopped by the operator. Normal program operation is defined as performing all four checkerboard patterns on all of available memory from every memory stack.

5. OPERATING PROCEDURE

5.1 Program and Operator Action

- a. Load the program into stack 0 using the procedure described in section 3.
- b. Set the SR to 200; press LOAD ADDRESS, and then start.
- c. The message "TEST LIMITS" will be printed. Specify the limits, via keyboard, as described in section 4.3.
- d. The message "SETUP SR" will be printed. Set the SR to 0000₍₈₎, and press the carriage return key.
- e. The program will perform all four tests on all of core memory specified, after which, automatic program relocation takes place.

5.2 Operational Switch Settings

Normal operation of the program requires the SR set to 0000₍₈₎. Refer to section 8.2, applications, for switch settings provided for trouble-shooting.

5.3 Subroutine Abstracts

5.3.1 The Patterns Used for Testing - The program executes a series of four tests on core memory. Each test writes a unique pattern, and checks each location for error.

Test 1 writes the value of each address into itself, from the lowest order to the highest order 4K field under test. The address pattern is then read and checked for error in the same direction, i.e., from the lowest to highest field under test. The pattern is then read and checked for error in the reverse directions, i.e., from the highest to the lowest field under test. When reading in the reverse direction, each location is repeatedly read a random number of times before reading the next location. The minimum number of reads per location is one, and the maximum is 20 octal. Error checking is done after each read.

Test 1 then writes the same address pattern again, only this time starting with the highest 4K field under test, and decrementing to the lowest under test. Reading and error checking is then performed as previously described.

Test 2 writes the ones complement value of each location into itself, and proceeds to read and check in the same manner as test 1.

Test 3 and 4 both write a "sliding" bit pattern throughout memory. The difference between the two tests being that test 3 rotates a single bit equal to 1, and test 4 rotates a single 0. Both tests use the following test sequence. The sequence is repeated 12 times, resulting in each bit of every memory location being complemented.

- a. Write all 1's into all of memory.
- b. Write a sliding 1 or 0 pattern into one 4K field.
- c. Read and test the 4K field in the forward direction only.
- d. Repeat steps b and c 11 more times before testing the next sequential field in the same manner.

After test 4 is completed on all memory fields, the program is relocated, and restarted with test 1.

6. ERRORS

Starting the program from address 200 will give a header print-out after the SR has been set up. The header identifies the information printed when a data error is found. The header appears as:

FIELD	OCTAL ADR.	GOOD	BAD	TEST
-------	------------	------	-----	------

Where: FIELD = an octal number (0 to 7) indicating the 4K field containing the error.

OCTAL ADR. = the memory address which contains the incorrect data.

GOOD = what the data in octal should have been.
BAD = the data as read. This will equal the good data except for one or more bits complemented.
TEST = the number (1 to 4) of the test which detected the error.

After each error print-out the program continues on with the next sequential memory location.

6.1 Error Halts and Description

Placing SR 0 on a 1 during an error print-out will cause a halt at location 2042. Press CONTINUE to resume testing.

7. RESTRICTIONS

7.1 Starting Restrictions

Start from address 200 to indicate the amount of core memory to test; to set up the SR and to receive a header print-out.

Starting from 207 requires the same operator action, but no header will be printed.

7.2 Operating Restrictions

None

8. MISCELLANEOUS

8.1 Execution Time

The time required to perform all four tests on one 4K memory stack is approximately 20 seconds.

8.2 Applications

For operating convenience, and as an aid to trouble-shooting, the SR may be used to control the program. The switch assignments and their effect on the program are described below. Please note that it is important that the program should be halted before changing the test selection switches. These switches are not sensed by the program during testing.

Halting the program with SR 0 is preferred, rather than with the STOP key. Using the STOP key may result in a halt while the program is in the process of relocating, which is disastrous.

8.2.1 Halt After Test or Error - SR 0 - Placing SR 0 on a 1 at any time while the program is running will cause a halt after the current test is completed. The MA will equal 2042 in the current stack containing the program. Press CONTINUE to resume testing, or restart from 200 or 207 to enter new parameters.

Placing SR 0 on a 1 during an error timeout will also cause a halt at location 2042. Proceed exactly as described in the above paragraph.

8.2.2 Inhibit Error Printout - SR 1 - Placing SR 1 on a 1 causes all error printouts to be inhibited. All other messages will not be inhibited. The program will continue to recognize errors, but will not print any information. SR 1 may be placed on a 1 or 0 while the program is running.

8.2.3 Bell on Error - SR 2 - SR 2 on a 1 causes the program to ring the TTY BELL whenever an error is detected. This is convenient when testing with power supply margins. SR 2 has precedence over SR 1 if both should happen to be on a 1. SR 2 may be placed on a 1 or 0 while the program is running.

8.2.4 Test Selection SR 3 Through 6 - Any one, or any combination of tests may be executed by placing any one or any combination of SR 3 through 6 on a 1. Test selections may be made only when starting from 200 or 207. SR 3 specifies test 1; SR 4, test 2; SR 5 test 3; SR 6 test 4. The test specified by the most significant SR on a 1 will be executed first.

If all four switches are on a 0, all four tests will be executed in order starting with test 1. Program relocation is not effected, regardless of the SR settings.

8.2.5 SR 7 and 8 - Not Used

8.2.6 Inhibit Program Relocation - SR 9 - The program normally relocates automatically as indicated by the INSTRUCTION FIELD indicators. To retain the program in its current 4K field, place SR 9 on a 1 at any time. Changing SR 9 to a 0 will permit relocation to resume.

8.2.7 SR 10 - Not used

8.2.8 Change Test Limits and SR - SR 11 - Placing SR 11 on a 1 will cause the program to automatically restart from address 207. The TEST LIMITS and SR may then be changed. SR 11 is sensed only after all specified tests have been completed on all of memory under test.

8.2.9 Loop on Address - A subroutine is provided which may be used to continuously loop on a single location, or a group of consecutive locations. No error checking is performed. The routine performs a read, and immediately follows with a write, on each location. The loop time between two reads, or two writes, is approximately 22.5 μ s.

Operating Procedure

- a. Set the INSTRUCTION FIELD switches to the current field, and the SR to 1137.
- b. Press LOAD ADDRESS
- c. Set the DATA FIELD switches to equal the 4K field number to test, and set the SR to equal the first address of the group.
- d. Press START. A halt will occur at 1142. Set the SR to equal the last address of the group.
- e. Press CONTINUE. The address (s) specified will be looped until stopped by the operator with STOP. SR 0 will not halt this routine.

To resume normal operation, restart the program from 200 or 207 of the current field.

9. PROGRAM DESCRIPTION

The Extended Memory address test is intended for use with a PDP-8, 8/I or 8/S equipped with the extended memory option. A total of four tests are executed by the program. Each test writes a unique pattern into core memory and then checks for error. The patterns were chosen so as to aid the operator in the event of addressing errors.

The program automatically relocates itself to each memory field under test to ensure that all fields may be correctly referenced from any field.

Control of the program is given to the operator by means of the SR. The operator may halt the program, inhibit error printouts, substitute the TTY BELL for error indication, halt after printout, select any one or a combination of the four tests, inhibit program relocation, and create an automatic program restart.

10. LISTING

/
 /PDP-8, 81, 8S EXTENDED MEMORY ADDRESS TEST.
 /START AT 200, RESTART AT 207 TO SKIP HEADER.
 /MIN. OF 8K OF CORE REQUIRED.
 /

0001	*1	
	/	
0001	JMP ,	
0002	0002	
0003	0003	
	/	
0010	*010	
	/	
6201	COF=6201	
6202	CIF=6202	
6214	RDF=6214	
6224	RIF=6224	
0010	0	
0011	0	
0012	0	
0013	0	
0014	0	
0015	0	
0016	0	
0017	0	
0020	0	
0021	0	
0022	1407	
0023	0000	DATAFLD, 0
0024	0026	FLAGS, 0
0025	0036	XLMTS, 0
0026	0026	INSFLD, 0
0027	7600	ERTBL, ERWRD
0030	7600	ENTBL, ERWRD+10
0031	7600	ERWRD, 7600
0032	7600	7600
0033	7600	7600
0034	7600	7600
0035	7600	7600
0036	7600	7600
0037	0000	MCWA, 0
0040	0010	K10, 10
0041	0740	K740, 740
0042	0400	K400, 400
0043	0200	K200, 200
0044	0100	K100, 100
0045	0040	K40, 40
0046	0020	K20, 20
0047	0276	XTST1, TST1
0050	0330	XTST2, TST2
0051	0400	XTST3, TST3
0052	0503	XTST4, TST4
0053	2200	XMOVE, CMOVE

4/1/68 16:50.4

XSETU, SETU1
K261, 261
K262, 262
K263, 263
K264, 264
M20, 7760

0054 1400
0055 0261
0056 0262
0057 0263
0060 0264
0061 7760

0062	7740	M40,	1740
0063	7774	M4,	1774
0064	7773	M5,	7773
0065	0000	TNUM,	0
0066	1066	XBANK,	CBANK
0067	1103	XTBANK,	NXTBANK
0070	0000	COUNT,	0
0071	0000	FLCNT,	0
0072	0000	LOOP,	0
0073	1200	XRROR,	ERROR
0074	0000	MEMADR,	0
0075	0000	FIRST1,	0
0076	0000	LAST1,	0
0077	6201	KCDF,	6201
0100	6202	KCIF,	6202
0101	2041	XHLT,	HALT
0102	0213	XRTN,	RTN1
0103	1125	XFILD,	FIELD
0104	1346	XPRER,	PRERR
0105	0007	K7,	7
0106	0000	CHAR,	0
0107	1674	XHDR,	PHDR
0110	1346	XPERR,	PRERR
0111	1315	XPING,	SPING

0112	7765	M13,	7765
0113	7764	M14,	7764
0114	7770	M10,	7770
0115	0260	K260,	260
0116	0215	K215,	215
0117	0377	K377,	377
0120	0370	K370,	370
0121	0277	K277,	277
0122	1354	XCRLF,	XCRLF
0123	0001	K1,	1
0124	0000	NXLOC,	0
0125	0000	EXIT,	0
0126	0000	PATR,	0
0127	0000	COMPR,	0
0130	0654	XWRF,	WFR1
0131	0720	XFWD,	RFM1
0132	1000	XBAB,	R8K1
0133	0671	XWRB,	WRB1
0134	0662	XWRF2,	WFR2
0135	0736	XFWD2,	RFM2
0136	1030	XBAB2,	R8K2
0137	0704	XWRB2,	WRB2
0140	1057	XONES,	ONES
0141	0000	SHIFT,	0
0142	7777	M1,	7777
0143	0000	DATA,	0

0200
0201
0202
0203
0204
0205
0206
0207
0210

*200
BEGIN,

IOF
CLA
RIF
DCA INSFLD
DCA FLAGS
JMS I XLMTS
JMS I XSTSR
JMS I XHDR
JMP RTN1

/PI OFF

/CLEAR PROGRAM FLAGS
/SETUP TEST LIMITS
/SETUP SR
/PRINT HEADER

/RESTART HERE
/

0211
0212
0213
0214
0215
0216
0217
0220
0221
0222
0223
0224
0225
0226
0227

RSTRT1,
RTN1,

ALAW,

JMS I XLMTS
JMS I XSTSR
RIF
DCA INSFLD
JMS I XFILD
TAD M10
DCA LOOP
7600
TAD ALAW
ISZ ERWRD
DCA I ERWRD
ISZ LOOP
JMP ALAW
TAD ERTBL
DCA ERWRD

/SET TEST LIMITS
/SETUP SR
/READ INSTRUCTION FIELD
/CURRENT FIELD

/-10

```

0230 1037 /EXAMINE SR      TAD MCWA
0231 0041      AND K740
0232 7440      SZA
0233 5237      JMP EXAM1
0234 1037      TAD MCWA
0235 1041      TAD K740
0236 3037      DCA MCWA
0237 7200      CLA
0240 1037      TAD MCWA
0241 0042      AND K400
0242 7440      SZA
0243 5447      JMP I XTST1
0244 7200      CLA
0245 1037      TAD MCWA
0246 0043      AND K200
0247 7440      SZA
0250 5450      JMP I XTST2
0251 7200      CLA
0252 1037      TAD MCWA
0253 0044      AND K100
0254 7440      SZA
0255 5451      JMP I XTST3
0256 7200      CLA
0257 1037      TAD MCWA
0260 0045      AND K40
0261 7440      SZA
0262 5452      JMP I XTST4
0263 4503      JMS I XFILD
0264 7604      LAS
0265 0123      AND K1
0266 7440      SZA
0267 5211      JMP RSTR1
0270 7604      LAS
0271 0046      AND K20
0272 7440      SZA
0273 5213      JMP RTN1
0274 5453      JMP I XMOVE
0275 2045      /XSTR, SETSR

```

/MADK 3,4,5 AND 6
 /JO ALL IF 0
 /SET ALL TEST BITS
 /SAVE
 /TEST 1 IF NO SKIP
 /TEST 2 IF NO SKIP
 /TEST 3 IF NO SKIP
 /TEST 4 IF NO SKIP
 /RESTORE DATA FIELD
 /CHECK SR 11
 /INHIBIT MOVE IF A 1
 /GO RELOCATE

/
/TEST 1, WRITE THE ADDRESS OF EACH LOCATION INTO
/ITSELF, THEN READ EACH STACK FORWARDS AND BACKWARDS,
/

0276 4454
0277 1055
0300 3065
0301 3074
0302 3126
0303 4466
0304 7410
0305 5313
0306 4530
0307 4531
0310 4532
0311 4467
0312 5301

TST1,
JMS I XSETU /SET DF TO 1ST FIELD
TAD K261 /TEST NUMBER
DCA TNUM
DCA MEMADR
DCA PATR
JMS I XBANK
SKP
JMP TST1A
JMS I XWRF /WRITE FORWARDS
JMS I XFWD /READ FORWARDS
JMS I XBANK /READ BACKWARDS
JMS I XTBNK /SETUP FOR NEXT FIELD
JMP TST1+3

/NOW WRITE BACKWARDS

0313 4454
0314 7240
0315 3074
0316 3126
0317 4466
0320 7410
0321 5244
0322 4533
0323 4531
0324 4532
0325 4467
0326 5314
0327 5214

TST1A,
JMS I XSETU /SET DF TO 1ST FIELD
CLA CMA
DCA MEMADR /SET ADR, COUNT TO 7777
DCA PATR /COMPARE WORD
JMS I XBANK /SEE IF FIELD HAS PROGRAM
SKP /BEGIN WRITING
JMP EXAM2 /NEXT TEST
JMS I XWRF /WRITE BACKWARDS
JMS I XFWD /READ FORWARDS
JMS I XBANK /READ BACKWARDS
JMS I XTBNK /SETUP FOR NEXT FIELD
JMP TST1A+1
JMP EXAM2 /NEXT TEST

```

/TEST 2: WRITE COMPLEMENT ADDRESS
/PATTERN FORWARD AND BACKWARD:
/
TST2,    JMS I XSETU    /SET OF TO 1ST FIELD
          TAD K262      /TEST NUMBER
          DCA TNUM
          DCA MEMADR    /SET ADDRESS COUNT TO 0
          CMA
          DCA PATR      /COMPARE WORD
          JMS I XBANK    /SEE IF FIELD HAS PROGRAM
          SKP           /WRITE
          JMP TST2A      /NEXT TEST
          JMS I XWRF2
          JMS I XFWD2    /READ FORWARDS
          JMS I XBAK2    /READ BACKWARDS
          JMS I XTBNK    /SETUP FOR NEXT FIELD
          JMP TST2+3
0330 4454
0331 1056
0332 3065
0333 3074
0334 7040
0335 3126
0336 4466
0337 7410
0340 5346
0341 4534
0342 4535
0343 4536
0344 4467
0345 5333

```

```

/NDW WRITE BACKWARDS
/
TST2A,   JMS I XSETU    /SET OF TO 1ST FIELD
          CLA CMA
          DCA MEMADR    /SET ADDRESS COUNT TO 777
          CMA
          DCA PATR      /COMPARE WORD
          JMS I XBANK    /SEE IF FIELD HAS PROGRAM
          SKP           /WRITE
          JMP EXAM3      /NEXT TEST
          JMS I XWRB2    /WRITE BACKWARDS
          JMS I XFWD2    /READ FORWARDS
          JMS I XBAK2    /READ BACKWARDS
          JMS I XTBNK    /SETUP FOR NEXT FIELD
          JMP TST2A+1
          JMP EXAM3      /NEXT TEST
0346 4454
0347 7240
0350 3074
0351 7040
0352 3126
0353 4466
0354 7410
0355 5251
0356 4537
0357 4535
0360 4536
0361 4467
0362 5347
0363 5251

```

/TEST3, WRITE ALL 1'S, THEN SLIDE A SINGLE 1 THRU EACH
/WORD, READ AND WRITE IN THE FORWARD DIRECTION ONLY.
/

0400 0400 0401 0402 0403 0404 0405 0406 0407 0410 0411 0412 0413 0414 0415 0416 0417 0420 0421 0422 0423 0424 0425 0426 0427 0430 0431 0432 0433 0434 0435

*400
/TST3,

JMS I XSETU /SET DF TO 1ST FIELD
TAD K263
DCA TNUM
IAC PATR /SET BIT 11 /COMPARE WORD
IAC /SAVE FIRST BIT
DCA COMPR /SET ADDRESS COUNT TO 0
DCA MEMADR /SEE IF FIELD HAS PROGRAM
JMS I XBANK /WRITE
SKP
JMP EXT3
JMS I XONES /WRITE 1'S
TAD M14 /-12 DECIMAL
DCA SHIFT
TAD PATR
DCA I MEMADR
ISZ MEMADR /DONE 4K WHEN SKIP
SKP /READ AND COMPARE
JMP RSL1 /CHECK FOR 12 POSITIONS
ISZ SHIFT /POSITION NEXT BIT
JMP ,+4 /START OVER WITH 11
TAD COMPR
DCA PATR
JMP WROT
TAD PATR
CLL RAL
SZL
IAC
DCA PATR /START NEXT WITH BIT 11
JMP WROT+2

WROT,

```

/READ AND TEST ROUTINE FOR SLIDING ONE
/
RSL1,      TAD COMPR      /FIRST POSITION
           DCA PATR
           DCA MEMADR      /SET ADDRESS COUNT TO 0
           TAD M14         /-12 DECIMAL
           DCA SHIFT
           TAD I MEMADR
           DCA DATA      /SAVE
           TAD DATA
           CIA             /2'S COMPLEMENT
           TAD PATR
           SEA CLA        /OK IF 0
           JMS I XRROR    /PRINT ERROR
           ISZ MEMADR     /DONE IF SKIP
           SKP CKDN1      /DONE
           JMP CKDN1      /CHECK FOR 12 POSITIONS
           ISZ SHIFT      /POSITION NEXT BIT
           JMP ,+4        /START OVER WITH 11
           TAD COMPR
           DCA PATH
           JMP RROT
           TAD PATR
           CLL RAL
           SZL
           IAC
           DCA PATR
           JMP RROT+2

```

```

0436 1127
0437 3126
0440 3074
0441 1113
0442 3141
0443 1474
0444 3143
0445 1143
0446 7041
0447 1126
0450 7640
0451 4473
0452 2074
0453 7410
0454 5270
0455 2141
0456 5262
0457 1127
0460 3126
0461 5241
0462 1126
0463 7104
0464 7430
0465 7001
0466 3126
0467 5243

```

```

/CKDN1,      TAD COMPR      /DONE ALL 12 IF NO SKIP
           CLL RAL
           SZL
           JMP ,+4
           DCA PATR
           TAD PATR
           JMP WROT=6      /WRITE IN SAME FIELD AGAIN
           JMS I XTBNK     /SETUP FOR NEXT FIELD
           JMP TST3+3
           JMP I ,+1      /NEXT TEST
           EXAM4

```

```

0470 1127
0471 7104
0472 7430
0473 5277
0474 3126
0475 1126
0476 5206
0477 4467
0500 5203
0501 5702
0502 0256

```

/TEST 4, WRITE ALL 1'S, THEN SLIDE A SINGLE 0 THRU EACH
 /WORD, WRITE AND READ IN THE FORWARD DIRECTION ONLY.

0503	4454	JMS I XSETU	/SET OF TO 1ST FIELD
0504	1060	TAD K264	/TEST NUMBER
0505	3065	DCA TNUM	
0506	7001	IAC	
0507	3127	DCA COMPR	/COMPARE WORD
0510	1127	TAD COMPR	
0511	7040	CMA	
0512	3126	DCA PATR	
0513	3074	DCA MEMADR	/SET ADDRESS COUNT TO 0
0514	4466	JMS I XBANK	/SEE IF FIELD HAS PROGRAM
0515	7410	SKP	/WRITE
0516	5751	JMP I XT4	
0517	4540	JMS I XONES	/WRITE 1'S
0520	7120	STL	/-12 DECIMAL
0521	1113	TAD M14	
0522	3141	DCA SHIFT	
0523	1126	TAD PATR	
0524	3474	DCA I MEMADR	
0525	2074	ISZ MEMADR	/DONE 4K WHEN SKIP
0526	7410	SKP	
0527	5752	JMP I XSL0	/READ AND COMPARE
0530	2141	ISZ SHIFT	/CHECK FOR 12 POSITIONS
0531	5336	JMP I+5	/POSITION NEXT BIT
0532	1127	TAD COMPR	/START OVER WITH BIT 11
0533	7040	CMA	
0534	3126	DCA PATR	
0535	5320	JMP WR1T	
0536	1126	TAD PATR	
0537	7004	RAL	
0540	7420	SNL	
0541	5344	JMP I+3	
0542	3126	DCA PATR	
0543	5323	JMP WR1T+3	
0544	7200	CLA	
0545	7001	IAC	
0546	7040	CMA	
0547	7120	STL	
0550	5342	JMP I+6	
0551	0263	EXAM4+5	
0552	0600	RSL0	

WRIT,

/ XT4,
XSL0,

```

0600 1127
0601 7040
0602 3126
0603 3074
0604 1113
0605 3141
0606 1474
0607 3143
0610 1143
0611 7040
0612 1126
0613 7040
0614 7640
0615 4473
0616 2074
0617 7410
0620 5242
0621 2141
0622 5227
0623 1127
0624 7040
0625 3126
0626 5204
0627 1126
0630 7120
0631 7004
0632 7420
0633 5236
0634 3126
0635 5206
0636 7200
0637 7001
0640 7040
0641 5234
0642 1127
0643 7104
0644 7420
0645 5652
0646 4467
0647 5653
0650 5651
0651 0263
0652 0507
0653 0506

*600
/READ AND TEST ROUTINE FOR SLIDING ZERO
/
RSL0, TAD COMPR /1ST POSITION
CMA
DCA PATR
DCA MEMADR /SET ADR, COUNT TO 0
TAD M14 /-12 DECIMAL
DCA SHIFT
TAD I MEMADR /READ
DCA DATA /SAVE
TAD DATA
CMA
TAD PATR
CMA
SZA CLA /MUST=0
JMS I XRROR /PRINT ERROR
ISZ MEMADR /DONE WHEN SKIP
SKP
JMP CKDN0
ISZ SHIFT /CHECK FOR 12 POSITIONS
JMP I+5 /START OVER WITH 11
TAD COMPR
CMA
DCA PATR
JMP RR1T
TAD PATR
STL
RAL
SNL
JMP I+3
DCA PATR
JMP RR1T+2
CLA
IAC
CMA
JMP I+5
/
CKDN0, TAD COMPR
CLL RAL
SNL
JMP I XRT4
/
JMS I XTBNK /SETUP FOR NEXT FIELD
JMP I XST4
JMP I I+1 /DONE ALL TESTS
EXAM4+5
/
XRT4, TST4+4
XST4, TST4+3

```

```

0654 0000 /
0655 1074 /WRITE FORWARD ROUTINE FOR TST1
0656 3474 /
0657 2074 /WRITE C(MEMADR) INTO
0660 5255 /SAME ADDRESS
0661 5654 /DONE WHEN SKIP
/EXIT

0662 0000 /
0663 1074 /WRITE FORWARD ROUTINE FOR TST2
0664 7040 /
0665 3474 /WRITE COMPLEMENT OF
0666 2074 /C(MEMADR) INTO SAME
0667 5263 /DONE 4K WHEN SKIP
0670 5662 /EXIT

0671 0000 /
0672 3072 /WRITE BACKWARD ROUTINE FOR TST1A
0673 1074 /
0674 3474 /4K COUNTER
0675 2072 /INITIALLY=7777
0676 7410 /DONE 4K WHEN SKIP
0677 5671 /EXIT
0700 1074 /SUBTRACT 1 FROM ADDRESS
0701 1142 /
0702 3074 /
0703 5273 /

0704 0000 /
0705 3072 /WRITE BACKWARDS ROUTINE FOR TST2A
0706 1074 /
0707 7040 /4K COUNTER
0710 3474 /INITIALLY=7777
0711 2072 /DONE 4K WHEN SKIP
0712 7410 /EXIT
0713 5704 /SUBTRACT 1 FROM ADDRESS
0714 1074 /
0715 1142 /
0716 3074 /
0717 5306 /

```

```

/READ FORWARD ROUTINE FOR TST1 AND TST1A
/
RFR1,
0
TAD I MEMADR
DCA DATA
TAD DATA
CIA
TAD PATR
SZA CLA
JMS I XRROR
ISZ MEMADR
SKP
JMP I RFR1
ISZ PATR
JMP RFR1+1
JMP I RFR1
/SAVE
/COMPARE WORD
/MUST EQUAL 0
/PRINT ERROR
/DONE 4K WHEN SKIP
/EXIT
/SHOULD NEVER GET HERE

```

```

0720 0000
0721 1474
0722 3143
0723 1143
0724 7041
0725 1126
0726 7640
0727 4473
0730 2074
0731 7410
0732 5720
0733 2126
0734 5321
0735 5720

```

```

/READ FORWARD ROUTINE FOR TST2 AND TST2A
/
RFR2,
0
TAD I MEMADR
DCA DATA
TAD DATA
CIA
TAD PATR
SZA CLA
JMS I XRROR
ISZ MEMADR
SKP
JMP I RFR2
TAD PATR
CIA
DCA PATR
JMP RFR2+1
/SAVE
/COMPARE WORD
/MUST=0
/PRINT ERROR
/DONE 4K WHEN SKIP
/EXIT
/SUBTRACT 1

```

```

0736 0000
0737 1474
0740 3143
0741 1143
0742 7041
0743 1126
0744 7640
0745 4473
0746 2074
0747 7410
0750 5736
0751 1126
0752 7041
0753 7040
0754 3126
0755 5337

```

```

1000 0000
1001 3072
1002 7040
1003 3074
1004 7040
1005 3126
1006 4665
1007 1474
1010 3143
1011 1143
1012 7041
1013 1126
1014 7640
1015 4473
1016 2071
1017 5207
1020 2072
1021 7410
1022 5600

1023 1074
1024 1142
1025 3074
1026 1074
1027 5205

*1000
/
/READ BACKWARD ROUTINE FOR IST1 AND IST1A
/
RBK1, 0
DCA LOOP /4K COUNTER
CMA
DCA MEMADR /SET ADR, COUNT TO 7777
CMA
DCA PATR /COMPARE WORD
JMS I XSALL
TAD I MEMADR /READ
DCA DATA /SAVE
TAD DATA
CIA
TAD PATR
SZA CLA /MUST=0
JMS I XRROR /PRINT ERROR
ISZ FLCNT /RANDOM LOOP
JMP LBK1 /DONE 4K WHEN SKIP
ISZ LOOP
SKP
JMP I RBK1 /EXIT

TAD MEMADR
TAD M1 /SUBTRACT 1 FROM ADDRESS
DCA MEMADR
TAD MEMADR
JMP LBK1=2

```

11030	0000
11031	3072
11032	7040
11033	3074
11034	3126
11035	4665
11036	1474
11037	3143
11040	1143
11041	7041
11042	1126
11043	7640
11044	4473
11045	2071
11046	5236
11047	2072
11050	7410
11051	5630
11052	2126
11053	1074
11054	1142
11055	3074
11056	5235
11057	0000
11060	7240
11061	3474
11062	2074
11063	5260
11064	5657
11065	2102

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PAUSE

/EXTENDED MEMORY ADDRESS TEST - TAPE 2
 /ROUTINE TO SEE IF TESTED FIELD HAS PROGRAM
 /

1066	0000	CBANK, 0	
1067	6224	RIF	/READ INST, FIELD
1070	3302	DCA SAVIF	/SAVE
1071	6214	RDF	/READ DATA FIELD
1072	7041	CIA	
1073	1302	TAD SAVIF	
1074	7640	SZA CLA	/EQUAL IF AC=0
1075	5666	JMP I CBANK	/DOESN'T HAVE PROGRAM
1076	4467	JMS I XTBNK	/INCREMENT DATA FIELD
1077	5666	JMP I CBANK	/TEST NEW FIELD
1100	2266	ISZ CBANK	/DONE ALL CAUSE PROGRAM NOW
		JMP I CBANK	/IN HIGHEST FIELD
1101	5666	SAVIF, 0	/EXIT
1102	0000		

Address	Instruction	Comment
1103	0000	
1104	7200	
1105	6214	
1106	7041	
1107	1076	
1110	7640	
1111	5314	
1112	2303	
1113	5321	
1114	6214	
1115	1040	
1116	1077	
1117	3320	
1120	6201	
1121	7604	
1122	7710	
1123	4501	
1124	5703	
1125	0000	
1126	7200	
1127	6214	
1130	3020	
1131	6224	
1132	1077	
1133	3334	
1134	6201	
1135	7200	
1136	5725	

```

1137 7200 /
1140 7604 /
1141 3075 /START HERE TO LOOP ON ADDRESS
1142 7402 /
1143 7604 /
1144 3076 /
1145 1075 /READ LOWER LIMIT
1146 3074 /NOW SETUP UPPER LIMIT
1147 1474 /
1150 3474 /
1151 1074 /
1152 7041 /
1153 1076 /
1154 7650 /
1155 5345 /
1156 2074 /
1157 5347 /
1160 7402 /

1161 0000 /
1162 7200 /
1163 1020 /
1164 1077 /
1165 3366 /
1166 6201 /
1167 7200 /
1170 5761 /

CLA /
LAS /
DCA FIRST1 /READ LOWER LIMIT
HLT /NOW SETUP UPPER LIMIT
LAS /
DCA LAST1 /
TAD FIRST1 /
DCA MEMADR /
TAD I MEMADR /READ
DCA I MEMADR /WRITE
TAD MEMADR /
CIA /
TAD LAST1 /
SNA CLA /
JMP OVER /
ISZ MEMADR /
JMP WRLOP /
HLT /

/ CFLO, /
CLA /
DCA LAST1 /
TAD KCDF /TEST FIELD
DCA :+1 /RESTORE TEST FIELD
CDF 00 /
CLA /
JMP I CFLO /EXIT

```



```

1254 4354 JMS CRLF /CP,LF
1255 1020 TAD DATFLD /TEST NUMBER
1256 7012 RTR
1257 7010 RAR
1260 1115 TAD K260
1261 4346 JMS PRERR /PRINT
1262 1113 TAD M14 /-12 DECIMAL
1263 3072 DCA LOOP
1264 4315 JMS SPING /SPACE 12
1265 1365 TAD OCADR /OCTAL ADR,
1266 3106 DCA CHAR /SAVE
1267 4326 JMS PROCTL /PRINT
1270 1114 TAD M10 /-8 DECIMAL
1271 3072 DCA LOOP
1272 4315 JMS SPING /SPACE 8
1273 1364 TAD GOOD
1274 3106 DCA CHAR
1275 4326 JMS PROCTL /PRINT
1276 1064 TAD M5 /-5
1277 3072 DCA LOOP /SPACE 5
1300 4315 JMS SPING
1301 1363 TAD BAD
1302 3106 DCA CHAR
1303 4326 JMS PROCTL /PRINT
1304 1064 TAD M5
1305 3072 DCA LOOP
1306 4315 JMS SPING /SPACE 5
1307 1065 TAD TNUM /TEST NUMBER
1310 4346 JMS PRERR /PRINT

```

```

1311 7604 / SW0,
1312 7710 SPA CLA /CH CK SR0
1313 4501 JMS I XHLT /GO HALT
1314 5246 JMP EREXT /EIT

/PRINT SPACES
/
SPING, 0 TAD K240 /SPACE
        TLS /PRINT
        TSF
        JMP '1
        ISZ LOOP
        JMP SPING+2
        CLA
        JMP I SPING /EXIT

```

```

1315 0000
1316 1370
1317 6046
1320 6041
1321 5320
1322 2072
1323 5317
1324 7200
1325 5715

```

```

1326 0000 /PRINT OCTAL
1327 1063 /
1330 3072 PROCTL, 0 TAD M4 /-4 /DIGIT COUNTER
1331 1106 DCA LOOP
1332 7104 POSITN, TAD CHAR
1333 7006 CLL RAL
1334 3106 RTL
1335 1106 DCA CHAR
1336 7004 TAD CHAR
1337 0105 RAL
1338 1115 AND K7
1341 4346 TAD K260 /MAKE ASCII
1342 2072 JMS PRERR /PRINT ONE
1343 5331 ISE LOOP
1344 7200 JMP POSITN /DO NEXT
1345 5726 CLA
JMP I PROCTL /EXIT

/PRINT A NUMBER
PRERR, 0
TLS
TSF
JMP I=-1
CLA
JMP I PRERR /EXIT

/CARRIAGE RETURN, LINE FEED
CRLF, 0
CLA
TAD K215 /CR
JMS PRERR /LF
TAD K212
JMS PRERR
JMP I CRLF

BAD, 0
GOOD, 0
OCADR, 0
LAST, 0
K207, 207
K240, 240
K212, 212

```

Address	Code	Label	Comment
1400	*1400		/ROUTINE TO SET DF TO FIRST TEST FIELD
1401	0000	SETU1, 0	
1402	7200	CLA	
1403	1075	TAD FIRST1	/FIRST TO TEST
1404	1077	TAD KCDF	
1405	3205	DCA ,+1	
1406	6201	CDF 00	/CHANGE TO TEST FIELD
1407	5600	JMP I SETU1	/EXIT
1408			/ROUTINE TO ACCEPT TEST LIMITS FROM
1409			/KEYBOARD INPUT
1410			/ROUTINE TO ACCEPT TEST LIMITS FROM
1411	0000	SLMTS, 0	
1412	4503	JMS I XFILD	
1413	4522	JMS I XCRLF	/CR, LF
1414	4751	JMS I XTLM	/PRINT TEST LIMITS
1415	4522	JMS I XCRLF	/CR, LF
1416	4303	JMS KEYIN	/GO ACCEPT INPUT
1417	4314	JMS LEGAL	/SEE IF IT'S LEGAL
1418	1106	TAD CHAR	
1419	1105	AND K7	
1420	7104	CLL RAL	/MASK AC 9-11
1421	7006	RTL	
1422	3075	DCA FIRST1	/POSITION TO AC 6-8
1423	4303	JMS KEYIN	/FIRST TO TEST
1424	1106	TAD CHAR	/WAIT FOR COMMA
1425	7041	CIA	/GET INPUT
1426	1353	TAD K254	
1427	7450	SNA	/OK IF 0
1428	5233	JMP ,+3	
1429	4344	JMS QUERY	/PRINT QUESTION MARK
1430	5211	JMP SLMTS+2	
1431	4303	JMS KEYIN	/WAIT FOR 2ND
1432	4303	JMS LEGAL	/SEE IF IT'S LEGAL
1433	4314	TAD CHAR	
1434	1106	AND K7	
1435	0105	CLL RAL	/MASK AC 9-11
1436	7104	RTL	
1437	7006	DCA LAST1	/POSITION TO AC 6-8
1438	3076	TAD FIRST1	/LAST TO TEST
1439	1075	CIA	
1440	7041	TAD LAST1	
1441	1076	SMA	/1ST IS > LAST IF NEG
1442	1075	JMP OKAS	
1443	7041	CLA	
1444	1076		
1445	7500		
1446	5256		
1447	7200		

```

1450 1075 TAD FIRST1
1451 3106 DCA CHAR
1452 1076 TAD LAST1
1453 3075 DCA FIRST1
1454 1106 TAD CHAR
1455 3076 DCA LAST1
1456 7200 CLA
1457 1076 TAD LAST1
1460 7041 CIA
1461 1075 TAD FIRST1
1462 7440 SZA
1463 5273 JMP ALOK
1464 1075 TAD FIRST1
1465 7041 CIA
1466 1023 TAD INSFLD
1467 7640 SZA CLA
1470 5273 JMP ALOK
1471 4752 JMS I XLCAT
1472 5211 JMP SLMTS+2
1473 4303 JMS KEYIN
1474 1106 TAD CHAR
1475 7041 CIA
1476 1116 TAD K215
1477 7450 SNA
1478 5607 JMP I SLMTS
1479 4344 JMS QUERY
1480 5211 JMP SLMTS+2

1503 0000 /
1504 6032 KEYIN,
1505 6031 KCC
1506 5305 KSF
1507 6036 JMP ,=1
1510 3106 KRB
1511 1106 DCA CHAR
1512 4510 TAD CHAR
1513 5703 JMS I XPERR
      JMP I KEYIN

```

OKAS,

ALOK,

KEYIN,

/LAST NOW IS FIRST
 /FIRST IS NOW LAST
 /SEE IF EQUAL
 /YES IF 0
 /NOW SEE IF IT HAS PROGRAM
 /CURRENT FIELD
 /NO IF A 1
 /PRINT PROGRAM LOCATION
 /AND START OVER
 /WAIT FOR C.R.
 /NOT A C.R, IF A SKIP
 /PRINT QUESTION MARK
 /START OVER

1514	0000	/	LEGAL,	0	TAD CHAR	
1515	1106				CIA	
1516	7041				TAD K377	/RUB-OUT IF 0
1517	1117				SNA CLA	
1520	7650				JMP SLMTS+2	
1521	5211				TAD CHAR	
1522	1106				AND K370	
1523	0120				CIA	
1524	7041				TAD K260	
1525	1115				SNA CLA	
1526	7650				JMP I LEGAL	
1527	5714				TAD CHAR	
1530	1106				CIA	/A CONNA IF 0
1531	7041				TAD K254	
1532	1353				SNA CLA	
1533	7650				JMP I LEGAL	
1534	5714				TAD CHAR	
1535	1106				CIA	
1536	7041				TAD K215	
1537	1116				SNA CLA	/A C.R, IF 0
1540	7650				JMP I LEGAL	
1541	5714				JMS QUERY	/QUERY
1542	4344				JMP SLMTS+2	/START OVER
1543	5211					
1544	0000	/	QUERY,	0	JMS I XCRLF	
1545	4522				TAD K277	
1546	1121				JMS I XPERR	/PRINT QUERY MARK
1547	4510				JMP I QUERY	
1550	5744					
1551	1646	/	XTLIM,		TLIMT	
1552	1600		XLCAT,		LOCAT	
1553	0254		K254,		254	

```

1600 /PRINT FIELD PROGRAM IS IN
1601 /
1602 *1600
1603 /
1604 LOCAT,
1605
1606 CLA
1607 TAD INSFLD /CURRENT FIELD
1608 RTR
1609 RAR
1610 AND K7
1611 TAD K260
1612 DCA FLDN
1613 TAD PRGAM
1614 DCA 12
1615 TAD I 12
1616 SNA
1617 JMP I LOCAT /DONE IF 0
1618 JMS I XPERR /EXIT
1619 JMP PLOCT /PRINT
1620
1621 /PRGAM,
1622
1623
1624
1625
1626
1627
1628
1629
1630
1631
1632
1633
1634
1635
1636
1637
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1639
1640
1641
1642
1643
1644
1645

```

320 /P
 322 /R
 317 /O
 307 /G
 322 /R
 301 /A
 315 /M
 240 /
 311 /I
 323 /S
 240 /
 311 /I
 316 /N
 240 /
 306 /F
 311 /I
 305 /E
 314 /L
 304 /D
 240 /
 0 /X
 0 /TERMINATOR

1600 0000
 1601 7200
 1602 1023
 1603 7012
 1604 7010
 1605 0105
 1606 1115
 1607 3244
 1610 1217
 1611 3012
 1612 1412
 1613 7450
 1614 5600
 1615 4510
 1616 5212
 1617 1617
 1620 0320
 1621 0322
 1622 0317
 1623 0307
 1624 0322
 1625 0301
 1626 0315
 1627 0240
 1630 0311
 1631 0323
 1632 0240
 1633 0311
 1634 0316
 1635 0240
 1636 0306
 1637 0311
 1640 0305
 1641 0314
 1642 0304
 1643 0240
 1644 0000
 1645 0000

```

1646 0000
1647 7200
1650 1257
1651 3012
1652 1412
1653 7450
1654 5646
1655 4510
1656 5252

1657 1657
1660 0324
1661 0305
1662 0323
1663 0324
1664 0240
1665 0314
1666 0311
1667 0315
1670 0311
1671 0324
1672 0323
1673 0000

/
TLIMIT, 0
CLA
TAD TSTL
DCA 12
PLIMIT, TAD I 12
SNA
JMS I TLIMIT
JMS I XPERR
JMP PLIMIT

/
TSTL, 324
/E
/S
/T
/
/L
/I
/M
/I
/T
/S
/TERMINATOR

/HEADER ROUTINE
/
PHDR, 0
JMS I XCRLF
TAD FILD
DCA 12
PFILD, TAD I 12
SNA
JMP ,+3
JMS I XPERR
JMP PFILD
TAD M5
DCA LOOP
JMS I XPING
TAD OTLDR
DCA 12

/PRINT TEST LIMITS
/DONE IF 0
/DONE IF 0
/CR, LF
/PRINT FIELD
/DONE IF 0
/SPACE 5

```

```

1712 1412      POCDR,  TAD I 12      /DONE IF 0      /PRINT OCTAL ADR
1713 7450      SNA
1714 5317      JMP ,+3
1715 4510      JMS I XPERR
1716 5312      JMP POCUR
/
1717 1064      TAD M5
1720 3072      DCA LOOP
1721 4511      JMS I XPING
1722 1355      TAD GODD
1723 3012      DCA 12
1724 1412      TAD I 12
1725 7450      SNA
1726 5731      JMP I ,+3
1727 4510      JMS I XPERR
1730 5324      JMP PG00D
1731 2000      BSPCE
PG00D,
/
1732 1732      FILD,
1733 0306      , 306
1734 0311      311
1735 0305      305
1736 0314      314
1737 0304      304
1740 0000      0
/
1741 1741      OTLDR,
1742 0317      , 317
1743 0303      303
1744 0324      324
1745 0301      301
1746 0314      314
1747 0240      240
1750 0301      301
1751 0304      304
1752 0322      322
1753 0256      256
1754 0000      0
/
1755 1755      GODD,
1756 0307      , 307
1757 0317      317
1760 0317      317
1761 0304      304
1762 0000      0
/
1763 5674      EXHDR,  JMP I PHDR

```

2000	/	*2000	
2001	/	BSPCE,	
2002		TAD M5	
2003		DCA LOOP	/SPACE 5
2004		JMS I XPING	
2005		TAD BADD	
2006		DCA 12	
2007		TAD I 12	/PRINT BAD
2008		SNA	/DONE IF 0
2009		JMP I+3	
2010		JMS I XPERR	
2011		JMP PBAD	
2012		TAD M5	
2013		DCA LOOP	
2014		JMS I XPING	/SPACE 5
2015		TAD TSTN	
2016		DCA 12	
2017		TAD I 12	/PRINT TEST
2018		SNA	/DONE IF 0
2019		JMP I+3	
2020		JMS I XPERR	
2021		JMP PTSTN	
2022		JMS I XCRLF	/CR, LF
2023		JMP I XPHDR	/EXIT
2024			
2025			
2026		TSTN,	
2027		324	/T
2028		305	/E
2029		323	/S
2030		324	/T
2031		0	
2032			
2033			
2034		BADD,	
2035		302	/B
2036		301	/A
2037		304	/D
2038		0	
2039			
2040			
2041		HALT,	
2042		0	
2043		HLT	
2044		JMP I HALT	/RESTART HEREOR RTN1
2045			
2046		XPHDR,	
2047		EXHDR	

```

2045 0000
2046 4503
2047 4522
2050 1270
2051 3012
2052 1412
2053 7450
2054 5257
2055 4510
2056 5252
2057 6036
2060 6031
2061 5265
2062 6036
2063 4510
2064 5645
2065 7604
2066 3037
2067 5260

2070 2070
2071 0323
2072 0305
2073 0324
2074 0325
2075 0320
2076 0240
2077 0323
2100 0322
2101 0000

2102 0000
2103 4503
2104 4317
2105 0316
2106 7040
2107 3071
2110 1020
2111 1077
2112 3313
2113 6201
2114 7200
2115 5702
2116 0017

/
/
/ WAIT HERE TO SETUP SR. TYPE CARRIAGE RETURN
/ AFTER SETTING SR,
/
SETSR, 0 JMS I XFILD /RESTORE DATA FIELD
JMS I XCRLF /CR, LF
TAD STSR
DCA 12
PSTSR, TAD I 12 /PRINT SETUP SR
SNA /DONE IF 0
JMP ,+3
JMS I XPERR
JMP PSTSR
KRB
KSF
JMP ,+4
KRB
JMS I XPERR
JMP I SETSR
LAS
DCA MCWA
JMP WTCR

/ STSR,
323 /S
305 /E
324 /T
325 /U
320 /P
240
323 /S
322 /R
0

/ STALL, 0 JMS I XFILD
JMS GENRAN /GET ANOTHER
AND K17
CMA
DCA FLCNT
TAD DATFLD
TAD KCDF
DCA ,+1
CDF 00
CLA
JMP I STALL /RESTORE DATA FIELD
K17, 0017 /EXIT
0017

```

2117	0000	GENRAN, 0	TAD RANTAB
2120	1355		CIA
2121	7041		TAD RANDEX
2122	1343		SZA CLA
2123	7640		JMP RANTAD-1
2124	5334		TAD TBLRAN
2125	1356		DCA RANDEX
2126	3343		TAD RANCON
2127	1342		CLL RAL
2130	7104		SZL
2131	7430		TAD K1
2132	1123		DCA RANCON
2133	3342		TAD I RANDEX
2134	1743		RANTAD, TAD RANCON
2135	1342		DCA I RANDEX
2136	3743		TAD I RANDEX
2137	1743		ISZ RANDEX
2140	2343		JMP I GENRAN
2141	5717	/	
2142	1234		RANCON, 1234
2143	2154		RANDEX, RANTBL+10
2144	4321		RANTBL, 4321
2145	1416		1416
2146	5363		5363
2147	6060		6060
2150	3035		3035
2151	2572		2572
2152	3237		3237
2153	0214		0214
2154	0000		0
2155	2154		RANTAB, 171
2156	2144		TBLRAN, RANTBL

```

/ROUTINE TO DETERMINE FIELD FOR RELOCATION
/
*2200
/
CMOVE,      JMS I XFILD      /SET DF TO CURRENT FIELD
2200 4503      7600
2201 7600      TAD ERTBL
2202 1024      DCA ERWRD      /SETUP ERROR TABLE POINTER
2203 3026      TAD FIRST1     /FIRST TESTED FIELD
2204 1075      CIA
2205 7041      TAD LAST1
2206 1076      SNA CLA        /LAST TESTED FIELD
2207 7650      JMP I XRTN     /DON'T MOVE IF EQUAL
2208 5502      TAD FLAGS      /START OVER
2209 1021      RAR
2210 7010      SZL            /FIRST MOVE IF A SKIP
2211 1021      JMP I XTMV     /SETUP FOR NEXT MOVE
2212 7430      IAC            /SET BIT 11
2213 5725      DCA FLAGS
2214 7001
2215 7001
2216 3021

2217 1076      TAD LAST1      /LAST TO TEST = 1ST MOVE
2218 3023      DCA INSFLD     /NEW CURRENT FIELD
2219 1023      TAD INSFLD
2220 1114      TAD M10        /SUBTRACT 1 FROM NEW CURRENT
2221 3124      DCA NXLOC     /NXLOC=DOEST'N FOR NEXT TIME
2222 6224      RIF
2223 7041      CIA
2224 1023      TAD INSFLD
2225 7650      SNA CLA        /IS NEXT SAME AS CURRENT
2226 5266      JMP SUB1      /YES, TRY NEXT LOWER FIELD
2227
2228
2229
2230

```

```

2231 2026 /CHECK FOR ERROR IN NEW FIELD
2232 1201 /
2233 7041 CKERR, ISZ ERWRD /POINTER+1
2234 1426 TAD CMOVE+1
2235 7650 CIA
2236 5310 TAD I ERWRD
/NO ERRORS RECORDED IF 0
/INITIALIZE MOVE

2237 1426 CNXT, TAD I ERWRD
2240 7041 CIA
2241 1023 TAD INSFLD
2242 7650 SNA CLA
2243 5253 JMP EQUAL
2244 1026 TAD ERWRD
2245 7041 CIA
2246 1025 TAD ENTBL
2247 7650 SNA CLA
2250 5310 JMP STMV
2251 2026 ISZ ERWRD
2252 5237 JMP CNXT
/ENTBL=ERWRD+10
/TABLE DONE IF 0
/INITIALIZE MOVE
/POINTER+1

/GET ERROR FIELD

2253 1426 TAD I ERWRD
2254 7041 CIA
2255 1075 TAD FIRST1
2256 7650 SNA CLA
2257 5502 JMP I XRTN
2260 1426 TAD I ERWRD
2261 7650 SNA CLA
2262 5266 JMP SUB1
2263 1023 TAD INSFLD
2264 1114 TAD M10
2265 3124 DCA NXLOC
/RESTORE TABLE POINTER

/ SUB1,
2266 1024 TAD ERTBL
2267 3026 DCA ERWRD
2270 1124 TAD NXLOC
2271 7041 CIA
2272 1023 TAD INSFLD
2273 7650 SNA CLA
2274 5253 JMP EQUAL
2275 1124 TAD NXLOC
2276 3023 DCA INSFLD
2277 1023 TAD INSFLD
2300 7041 CIA
2301 1075 TAD FIRST1
2302 7650 SNA CLA
2303 5231 JMP CKERR
2304 1023 TAD INSFLD
2305 1114 TAD M10
2306 3124 DCA NXLOC
2307 5231 JMP CKERR
/NEXT = CURRENT NEXT IF 0
/NEW CURRENT FIELD
/IS IT = LOWEST FIELD
/YES
/CURRENT NEW FIELD
/SUBTRACT 1 FROM DF
/NEXT FIELD LOWER

```

2310	7200	CLA			
2311	1024	TAD	ERTBL		
2312	3026	DCA	ERWRD		/RESTORE TABLE POINTER
2313	6224	RIF			
2314	3723	DCA	I XSRCE		
2315	1723	TAD	I XSRCE		
2316	7041	CIA			
2317	1023	TAD	INSFLD		
2320	7650	SNA	CLA		/DON'T MOVE IF EQUAL
2321	5502	JMP	I XRTN		/START OVER
2322	5724	JMP	I XMVE		/GO MOVE
2323	2522	SOURCE			
2324	2507	MOVE			
2325	2400	NXTMV			

/ STMV,

/ XSRCE,
XMVE,
XTMV,
/

[illegible]

```

2455 1024      /SUB2,
2456 3026      TAD ERTBL
2457 1124      DCA ERWRD
2460 7450      TAD NXLOC
2461 5502      SNA I XRTN
2462 1114      TAD M10
2463 3124      DCA NXLOC
2464 1124      TAD NXLOC
2465 7041      CIA
2466 1023      TAD INSFLD
2467 7640      SZA CLA
2470 5203      JMP CHNXT
2471 1124      TAD NXLOC
2472 7450      SNA
2473 5203      JMP CHNXT
2474 5262      JMP SUB2+5

2475 1076      /NXTHI,
2476 3124      TAD LAST1
2477 1076      DCA NXLOC
2478 3023      TAD LAST1
2479 5203      DCA INSFLD
2480 5203      JMP CHNXT

2502 1124      /MVBK,
2503 3023      TAD NXLOC
2504 6224      DCA INSFLD
2505 3322      RIF
2506 3021      DCA SOURCE
                DCA FLAGS

```

```

/RESTORE TABLE POINTER
/NEXT LOWER FIELD
/=FIELD 0 IF 0
/START OVER CAN'T MOVE
/SUBTRACT 1
/NOW = 2 FIELDS LOWER

/CURRENT FIELD
/ARE THEY EQUAL
/NO
/YES
/DOES IT = FIELD 0
/YES
/NO

/VERY LAST TO TEST
/MAKE IT NEXT FIELD

```

/CLEAR BIT 11

```

2507 1077 /ROUTINE TO RELOCATE 4K FIELDS
2510 1322 /
2511 3322 /MOVE, TAD KCDF /6201
2512 1077 TAD SOURCE /CURRENT FIELD
2513 1023 DCA SOURCE /SOURCE NOW = CDF N
2514 3324 TAD KCDF /6201
2515 7040 TAD INSFLD /NEW FIELD
2516 3010 DCA DESTN /DESTN NOW = CDF N
2517 7040 CMA /SOURCE COUNT
2520 3011 DCA 10 /DESTINATION COUNT
2521 3072 DCA 11 /4K COUNTER
2522 0000 DCA LOOP /WILL = CDF N
2523 1410 TAD I 10 /TAKE FROM HERE
2524 0000 DESTN, 0 /PUT IN HERE
2525 3411 DCA I 11 /DONE 4K WHEN SKIP
2526 2072 ISZ LOOP /KEEP MOVING
2527 5322 JMP SOURCE /6202
2530 1100 TAD KCIF /NEW FIELD
2531 1023 TAD INSFLD /CHANGE TO NEW FIELD
2532 3333 DCA I+1 /EXIT TO RTN1 IN
2533 6202 CIF 00 /NEW FIELD
2534 5502 JMP I XRTN

```

\$

THERE ARE NO ERRORS

SYMBOL TABLE

ALAW	0220
ALOK	1473
BAD	1363
BADD	2034
BEGIN	0200
BSPCE	2000
CBANK	1066
CDF	6201
CFLD	1161
CHAR	0106
CHNXT	2403
CIF	6202
CKDN0	0642
CKDN1	0470
CKERR	2231
CKNT	2442
CKNXT	2411
CMOVE	2200
CNXT	2237
COMPR	0127
COUNT	0070
CRLF	1354
DATA	0143
DATFLD	0020
DESTN	2524
ENTBL	0025
EPRNT	1254
EQUAL	2253
EREXT	1246
ERROR	1200
ERTBL	0024
ERWRD	0026
EXAM1	0237
EXAM2	0244
EXAM3	0251
EXAM4	0256
EXHDR	1763
EXIT	0125
EXT3	0501
EXT4	0650
FIELD	1125
FILD	1732
FIRST1	0075
FLAGS	0021
FLCNT	0071
FLDN	1644
GENRAN	2117
GDD	1755
GOOD	1364
HALT	2041
INSFLD	0023
KCDF	0077
KCIF	0100

SYMBOL TABLE

KEYIN	1503
K1	0123
K10	0040
K100	0044
K17	2116
K20	0046
K200	0043
K207	1367
K212	1371
K215	0116
K240	1370
K254	1553
K260	0115
K261	0055
K262	0056
K263	0057
K264	0060
K277	0121
K370	0120
K377	0117
K40	0045
K400	0042
K7	0105
K740	0041
LAST	1366
LAST1	0076
LBK1	1007
LBK2	1036
LEGAL	1514
LOCAT	1600
LOOP	0072
MC'A	0037
MEMADR	0074
MOVE	2507
MVBK	2502
M1	0142
M10	0114
M13	0112
M14	0113
M20	0061
M4	0063
M40	0062
M5	0064
NXLOC	0124
NXTBnk	1103
NXTHI	2475
NXTMV	2400
OCADR	1365
OKAS	1456
ONES	1057
OTLDR	1741
OVER	1145
PATR	0126

SYMBOL TABLE

PBAD	2005
PFILD	1700
PGOOD	1724
PHDR	1674
PLIMT	1652
POCT	1612
POCDR	1712
POSITN	1331
PRERR	1346
PRGAM	1617
PROCTL	1326
PSTSR	2052
PTSTN	2017
QUERY	1544
RANCON	2142
RANDEX	2143
RANTAB	2155
RANTAD	2135
RANTBL	2144
RBK1	1000
RBK2	1030
RDF	6214
RFR1	0720
RFR2	0736
RIF	6224
RROT	0441
RR1T	0604
RSL0	0600
RSL1	0436
RSTRT1	0211
RTN1	0213
SAVIF	1102
SETSR	2045
SETU1	1400
SHIFT	0141
SLMTS	1407
SOURCE	2522
SPING	1315
STALL	2102
STMV	2310
STNX	2447
STNXT	2425
STSR	2070
SUB1	2266
SUB2	2455
SW0	1311
SW1	1242
SW2	1233
TBLRAN	2156
TLIMT	1646
TNUM	0065
TSTL	1657
TSTN	2026

SYMBOL TABLE

TST1	0276
TST1A	0313
TST2	0330
TST2A	0346
TST3	0400
TST4	0503
WFR1	0654
WFR2	0662
WRB1	0671
WRB2	0704
WROL	1147
WROT	0414
WR1T	0520
WTCR	2060
XBAK	0132
XBAK2	0136
XBANK	0066
XCRLF	0122
XFILD	0103
XFWD	0131
XFWD2	0135
XHDR	0107
XHLT	0101
XLCA	1552
XLWTS	0022
XMOVE	0053
XMOVE	2324
XONES	0140
XPERR	0110
XPHDR	2044
XPING	0111
XPRER	0104
XRROR	0073
XRTN	0102
XRT4	0652
XSALL	1065
XSETU	0054
XSL0	0552
XSRCE	2323
XSTSR	0275
XST4	0653
XIBNK	0067
XTLIM	1551
XTMV	2325
XTST1	0047
XTST2	0050
XTST3	0051
XTST4	0052
XT4	0551
XWRB	0133
XWRB2	0137
XWRF	0130
XWRF2	0134

SYMBOL TABLE

SYMBOL TABLE

DATFLD	0020
FLAGS	0021
XLMTS	0022
INSFLD	0023
ERTBL	0024
ENTBL	0025
ERWRD	0026
MCWA	0037
K10	0040
K740	0041
K400	0042
K200	0043
K100	0044
K40	0045
K20	0046
XTST1	0047
XTST2	0050
XTST3	0051
XTST4	0052
XMOVE	0053
XSETU	0054
K261	0055
K262	0056
K263	0057
K264	0060
M20	0061
M40	0062
M4	0063
M5	0064
TNUM	0065
XBANK	0066
XTBNK	0067
COUNT	0070
FLCNT	0071
LOOP	0072
XRROR	0073
MEMADR	0074
FIRST1	0075
LAST1	0076
KCDF	0077
KCIF	0100
XHLT	0101
XRTN	0102
XFILD	0103
XPRER	0104
K7	0105
CHAR	0106
XHDR	0107
XPERR	0110
XPING	0111
M13	0112
M14	0113
M10	0114

SYMBOL TABLE

K260	0115
K215	0116
K377	0117
K370	0120
K277	0121
XCRLF	0122
K1	0123
NXLOC	0124
EXIT	0125
PATR	0126
COMPR	0127
XWRF	0130
XFWD	0131
XBAK	0132
XWRB	0133
XWRF2	0134
XFWD2	0135
XBAK2	0136
XWRB2	0137
XONES	0140
SHIFT	0141
M1	0142
DATA	0143
BEGIN	0200
RSTRT1	0211
RTN1	0213
ALAW	0220
EXAM1	0237
EXAM2	0244
EXAM3	0251
EXAM4	0256
XSTSR	0275
TST1	0276
TST1A	0313
TST2	0330
TST2A	0346
TST3	0400
WROT	0414
RSL1	0436
RROT	0441
CKDN1	0470
EXT3	0501
TST4	0503
WR1T	0520
XT4	0551
XSL0	0552
RSL0	0600
RR1T	0604
CKDN0	0642
EXT4	0650
XRT4	0652
XST4	0653
WFR1	0654

SYMBOL TABLE

WFR2	0662
WRB1	0671
WRB2	0704
RFR1	0720
RFR2	0736
RBK1	1000
LBK1	1007
RBK2	1030
LBK2	1036
ONES	1057
XSALL	1065
CBANK	1066
SAVIF	1102
NXTBANK	1103
FIELD	1125
OVER	1145
WRLOP	1147
CFLO	1161
ERROR	1200
SW2	1233
SW1	1242
EREXT	1246
EPRNT	1254
SW0	1311
SPING	1315
PROCTL	1326
POSITN	1331
PRERR	1346
CRLF	1354
BAD	1363
GOOD	1364
OCADR	1365
LAST	1366
K207	1367
K240	1370
K212	1371
SETU1	1400
SLMTS	1407
OKAS	1456
ALOK	1473
KEYIN	1503
LEGAL	1514
QUERY	1544
XTLIM	1551
XLCAT	1552
K254	1553
LOCAT	1600
PLOCT	1612
PRGAM	1617
FLON	1644
TLIMT	1646
PLIMT	1652
TSTL	1657

SYMBOL TABLE

PHDR	1674
PFILN	1700
POCDR	1712
PGOOD	1724
FILD	1732
OTLDR	1741
GODD	1755
EXHDR	1763
BSPCE	2000
PBAD	2005
PTSTN	2017
TSTN	2026
BADD	2034
HALT	2041
XPHDR	2044
SETSR	2045
PSTSR	2052
WTCR	2060
STSR	2070
STALL	2102
17	2116
KENRAN	2117
RANTAD	2135
RANCON	2142
RANDEX	2143
RANTBL	2144
RANTAB	2155
TBLRAN	2156
CMOVE	2200
CKERR	2231
CNXT	2237
EQUAL	2253
SUB1	2266
STMV	2310
XSRCE	2323
XMVE	2324
XTMV	2325
NXTMV	2400
CHNXT	2403
CKNXT	2411
STNXT	2425
CKNT	2442
STNX	2447
SUB2	2455
NXTHI	2475
MVBK	2502
MOVE	2507
SOURCE	2522
DESTN	2524
CDP	6201
CIF	6202
RUF	6214
RIF	6224

SYMBOL TABLE